Lab 1: Requirements Engineering & UML Use-Case Modelling

NAME: Shiva nandan R

SECTION: H

SRN: PES2UG24CS468

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-001	Function	Multi-Vendor Split Payment	High	**Pass:** Split payouts to all vendors correctly balance with the total cart value after 5% commission deduction. **Fail:** Any vendor payout or total calculation is incorrect.	Ensures vendors receive their correct earnings after commission	Core feature for fair and automated vendor payments.
FR-002	Function	Vendor Storefront & Product Management	High	**Pass:** Vendors can successfully add, update, and delete products with valid details. **Fail:** Product information is not saved, updated, or deleted correctly.	Allows artisans to independently manage their products and stores.	Essential for independent vendor operations.
FR-003	Function	Shopping Cart & Order Management	High	**Pass:** Shoppers can add products from multiple vendors, modify quantities, and successfully	Enables customers to conveniently purchase products from multiple vendors.	Required for the basic shopping and purchasing process.

				place an order. **Fail:** Cart items, quantities, or order details are lost or calculated incorrectly		
FR-004	Function	Order Tracking & Status Management	Medium	**Pass:** Shoppers can view the latest order status and vendors can update their assigned orders through valid statuses. **Fail:** Status updates are missing, incorrect, or not visible to the shopper.	Provides transparency on order progress for shoppers and vendors.	Improves order visibility and customer experience.
FR-005	Function	Product Search & Filtering	Medium	**Pass:** Search and filters return products matching the selected keyword, category, price, vendor, rating, or availability. **Fail:** Results contain irrelevant products or omit matching products.	Helps shoppers quickly find relevant products.	Makes product discovery faster and easier.
NFR-001	Non-Function	Fast & Secure Product Delivery	High	**Pass:** High-resolution product images load in **<500 ms** under simulated peak load while meeting	Ensures fast product browsing while protecting the platform from security risks.	Important for performance and secure user experience.

				defined security standards. **Fail:** Response time exceeds 500 ms or security requirements are violated.		
NFR-002	Non-Function	Security & Reliability	High	**Pass:** User data and payment information are securely transmitted and protected, and the system remains operational during concurrent transactions. **Fail:** Unauthorized access, data exposure, insecure transmission, or service failure occurs	Protects sensitive user/payment data and ensures dependable transactions.	Critical for protecting data and ensuring reliable transactions.

Use-Case Flow Document

Field	Details
Goal	Place Order
Actor	Shopper with help from Payment Gateway and Artisan Vendor
Main flow	**1.** Shopper selects products from one or more artisan vendors and proceeds to **checkout**. **2.** The system displays the selected products, quantities, vendor details, delivery information, and total cart value. **3.** Shopper reviews the order and confirms the delivery details.

	4. The system calculates the total order amount. **5.** Shopper selects a payment method and submits the payment. **6.** The system sends the payment request to the **Payment Gateway** for authorization. **7.** The Payment Gateway successfully authorizes the payment. **8.** The system deducts the **5% platform commission** from the applicable vendor earnings. **9.** The system splits the remaining payment among the respective **Artisan Vendors** based on their items in the order. **10.** The system creates the order and assigns the relevant items to each vendor. **11.** The system displays the order confirmation and order summary to the shopper. **12.** The checkout process is completed.
Alternate Flow	**Payment Declined** At **Step 7**, if the payment is declined: **7a.** The system displays a payment failure message to the shopper. **7b.** No vendor payout or commission deduction is performed. **7c.** Shopper may retry using the same or a different payment method. **7d.** If payment continues to fail, the order is not created.

Use Case Model

